

Muhammad Mujeeb Ur Rahman

AI Engineer / Software Engineer



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[Portfolio website](#)



[Github](#)

About Me

AI Engineer and Software Engineering student with hands-on experience in Computer Vision, Deep Learning, Machine Learning and Generative AI. Expert with Python, TensorFlow, PyTorch, OpenCV, YOLO, designing intelligent systems for object detection, facial recognition, crowd analysis, medical image classification and real-time computer vision applications. Proficient in creating LLM-powered agents and end-to-end AI applications, from recommendation systems, NLP models to RAG pipelines from start to finish, with LangChain, LangGraph, vector databases, and workflow automation with n8n. Experience in industry comprises of an internship in Generative and Agentic AI at NCAI Peshawar, an AI/CV Engineer position with Zentify Systems, and internships in Machine Learning / data science for real world model development.

Work Experience

Computer Vision Engineer | Zentify Systems (in collaboration with Ertikaaz AI-Mutaadid Co., KSA) | 2026 - Present

- Developing a computer vision-based Smart Parking Management System using YOLO, ByteTrack, and EasyOCR for real-time vehicle detection, tracking, and license plate recognition.
- Designed a two-camera pipeline architecture to handle end-to-end vehicle flow, from entrance detection to parking-slot verification.
- Working directly with real-world CCTV footage, addressing practical deployment challenges for a production-track, multi-camera system.

AI/ML Engineer Intern | ITSOLERA | Jan 2026 - Apr 2026

- Contributed to AI and Machine Learning projects involving data preprocessing, exploratory analysis, model development, and evaluation.
- Developed and optimized machine learning solutions using Python and relevant AI/ML libraries.
- Assisted in model testing, debugging, and performance enhancement for real-world applications.
- Collaborated with cross-functional teams to support AI-driven decision-making and project delivery.
- Demonstrated strong analytical, problem-solving, and research skills while adapting to emerging technologies and project requirements.

AI Engineer Intern | NCAI Peshawar | Dec 2025 - Feb 2026

- Worked on AI-based projects using LLMs, LangChain, and RAG pipelines.
- Built agents and automation workflows using n8n for chatbots and backend processes.
- Developed and improved AI chat systems and smart assistants.
- Gained practical experience in APIs, vector databases, and real-world AI applications.

Research Papers

- Muhammad Mujeeb Ur Rahman et al., "MemAdmit: Epistemic Gating and Evidence-Calibrated Memory Admission for LLM Agents," submitted to IEEE Transactions on Artificial Intelligence, 2026.

Education

Bachelor of Software Engineering | University of Malakand | 2023 - 2027

Projects

CampusMind — AI Study Mentor

Built an AI-powered study assistant that generates summaries, flashcards, MCQs, viva questions, and personalized study plans from uploaded material. Designed the architecture connecting a React frontend to a Node.js/Express backend, orchestrated through n8n workflows calling the Gemini API, with the system deployed via Vercel and n8n Cloud.

AI Employee OS

Building the AI layer of a multi-tenant SaaS platform where AI agents (Sales, HR, Finance, Support) perform real business actions through CRM, invoicing, and email integrations rather than simple chat responses. Designed and implemented a router/delegate-agent architecture across 49 tools and 13 agents, including AI-powered reporting, Stripe payment automation, OCR, and meeting transcription, with full tenant isolation and a 176-test automated suite validating correctness and security.

MeetingMind — AI Meeting Intelligence System

Developing the backend for an AI system that converts meeting recordings into structured summaries, speaker-wise transcripts, decisions, and action items with deadlines. Built a FastAPI backend integrating DeepGram for speech-to-text/diarization and Gemini for summarization, decision extraction, and contextual Q&A, with async SQLAlchemy handling a 14-table relational schema.

Blog Writing Agent

Developed a multi-agent AI content generation system using LangGraph, structured around a Router–Orchestrator–Workers–Reducer pipeline that plans, researches (via Tavily), writes sections in parallel, and integrates AI-generated images into a final blog post. Built with Google Gemini for generation and a real-time Streamlit interface, with exportable Markdown/ZIP output.

FashionHub — AI Fashion Sales Assistant

Contributed to an AI-powered sales assistant integrating WhatsApp and Instagram, reviewing and verifying an AI coding agent's implementation across a FastAPI/LangGraph AI service, Node.js backend, and React admin dashboard. Identified and corrected multiple false "completion" claims from the coding agent through direct code and log verification, ensuring the delivered system was genuinely tested before handoff.

Adaptive Synthetic Data Augmentation Toolkit

Built a machine learning pipeline to address class imbalance in fraud detection by generating targeted synthetic data for underrepresented cases. Trained and evaluated classification models before and after augmentation, improving fraud-class recall from 72% to 76% and F1-score from 0.83 to 0.85, and deployed the model through an interactive Streamlit application.

Skills

Computer Vision: Object Detection, Crowd Detection, Face Recognition, Emotion Detection, Medical Image Analysis, Video Analytics

LLM & AI Frameworks: LLMs, LangChain, LangGraph, RAG Pipelines, Vector Databases, AI Agents, n8n Automation

Deep Learning & ML: CNNs, TensorFlow, Keras, PyTorch, Scikit-Learn, Transfer Learning

Programming: Python, C++, JavaScript

Tools: OpenCV, YOLO, Git, GitHub, Jupyter Notebook, Streamlit

Databases: PostgreSQL, MongoDB, Firebase, ChromaDB, Pinecone

Languages

- Urdu | Native Proficiency
- Pashto | Native Proficiency
- English | Professional Working Proficiency
- Arabic | Professional Working Proficiency